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► To cite this version:

Rémy Vallot, Thibault Dairay, Florian de Vuyst, Mathilde Mougeot. Convergence acceleration of a nonlinear solver using statistical learning. Application of Digital Twins to Large-Scale Complex Systems, Dec 2025, Illinois, United States. , 2025. hal-05449018

HAL Id: hal-05449018

<https://hal.science/hal-05449018v1>

Submitted on 8 Jan 2026

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CONVERGENCE ACCELERATION OF A NONLINEAR SOLVER USING STATISTICAL LEARNING

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Application of Digital Twins to Large-Scale Complex Systems, Chicago, December 1 — 5, 2025

Introduction

- Numerical simulations in industry : crucial to help decision making.
- Traditional iterative algorithms : convergence heavily dependent on initialization.
- This work explores alternative methods to improve initialization, accelerating convergence while preserving physical accuracy.

Problem definition

Nonlinear PDE → high-dimensional nonlinear algebraic equations, solved using a Quasi-Newton method.

Finding U^* such that $F(U^*, \lambda) = 0$
where λ is a vector of parameters.

Applications

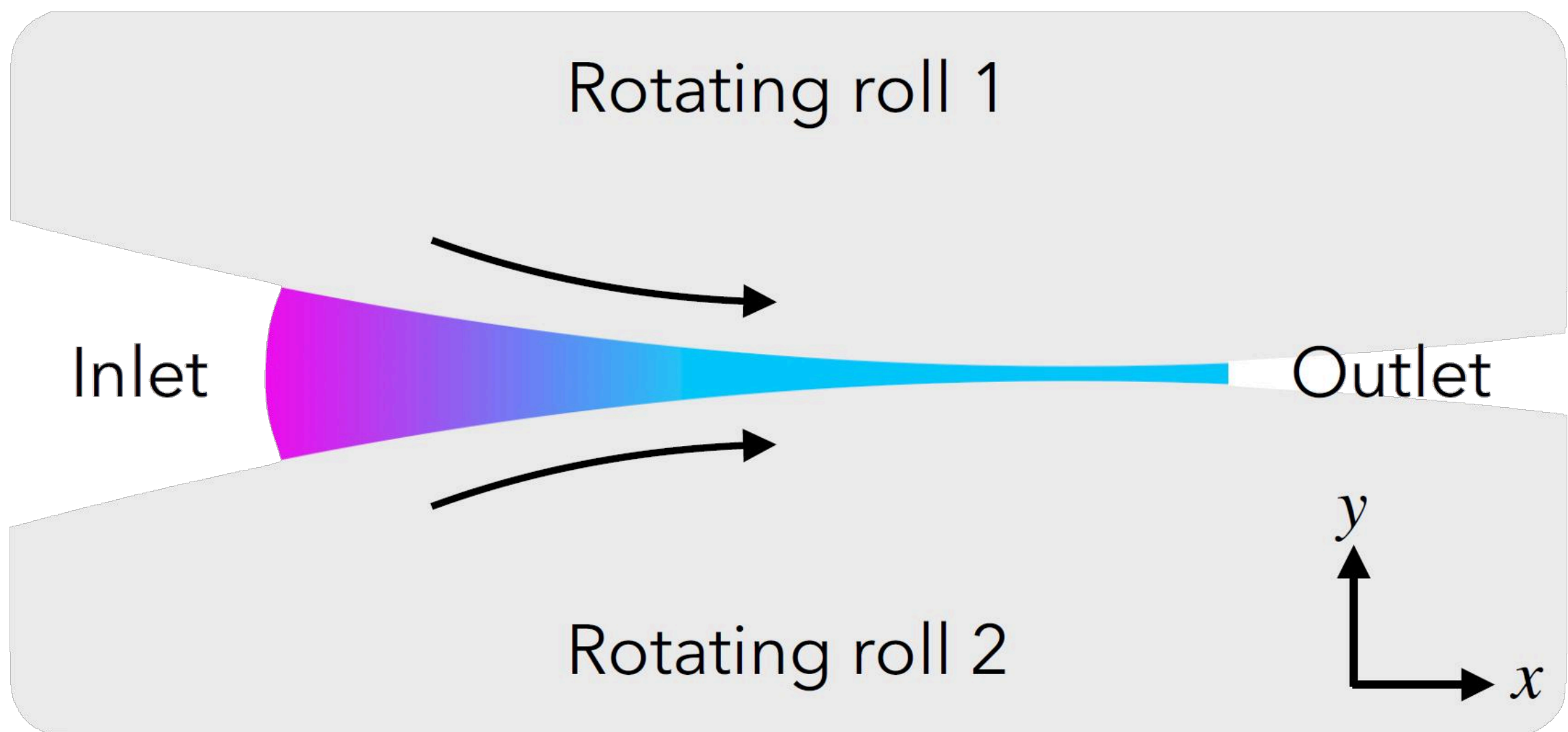
- 1D Nonlinear Poisson equation

Let Ω be an open domain of $\mathbb{R}$.
The nonlinear Poisson equation is formalised as follows:

$$\begin{cases} -\frac{\partial}{\partial x} \left[q(u) \frac{\partial}{\partial x} u \right] = g(x, \lambda) & \text{in } \Omega \\ u = u_D & \text{on } \partial\Omega \end{cases}.$$

- 2D calendering process

Rubber process manufacturing : compress material between two counter-rotating rolls.
Focus on mechanical fields. Heat transfers are not considered in this study.



Governing equations:

Mass and momentum conservation for incompressible highly viscous flow : divergence free velocity and nonlinear Stokes equations.
Nonlinear power-law viscosity model : λ contains the coefficients of the viscosity model.

Strategy

Use machine learning to find a new initialization for the nonlinear solver generating less iterations to converge.

Data & baselines

Dataset:
a set of vectors $\{\lambda_k\}$ of parameters
a set of associated discretized solutions $\{U_k^*\}$ computed using the solver.

Sampling strategy:

Train: regular grid 10×10 in the parameter space.
Test: 100 pairs via Latin Hypercube Sampling in the same intervals.

2 baseline strategies:

Classical initialization: constant field set to boundary value.

Nearest initialization: use the solution from the closest parameter in the database.

Tested methods

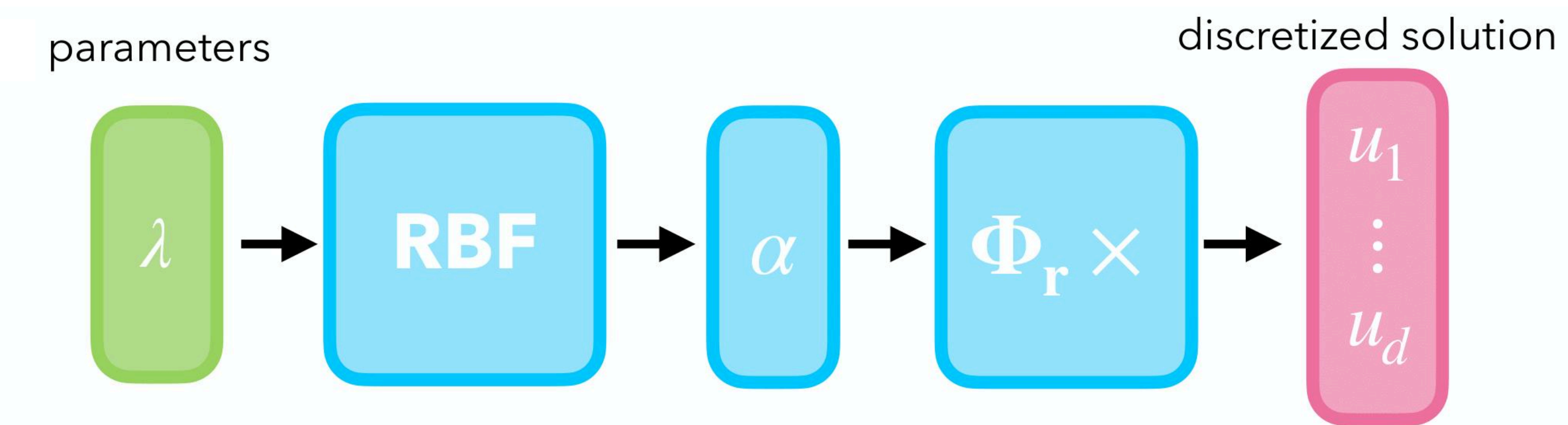
All model diagrams follow this color scheme



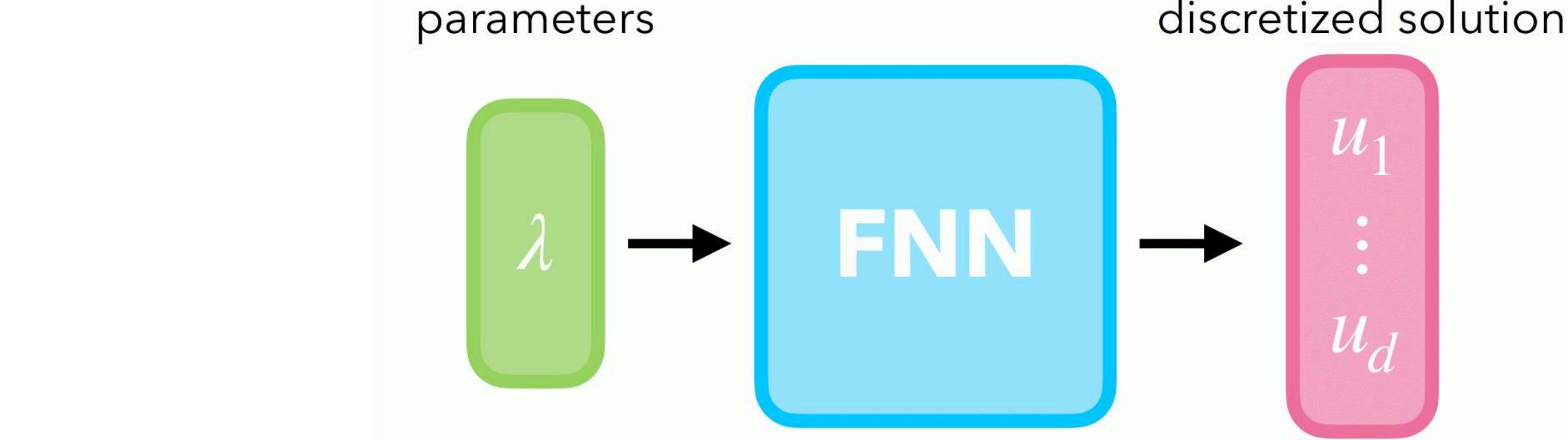
- Proper Orthogonal Decomposition (POD) [4]

Consider a snapshot matrix $U = \begin{bmatrix} | & | & & | \\ U_1^* & U_2^* & \dots & U_n^* \\ | & | & & | \end{bmatrix}$.

Compute truncated SVD ($U = \Phi_r \Sigma_r V_r^T$) to obtain Φ_r , the orthonormal reduced basis between projection coefficients α_k and solution U_k .
Define interpolation method like Radial Basis Function (RBF) between parameters and projection coefficients.



- Fully connected Neural Networks (FNN)



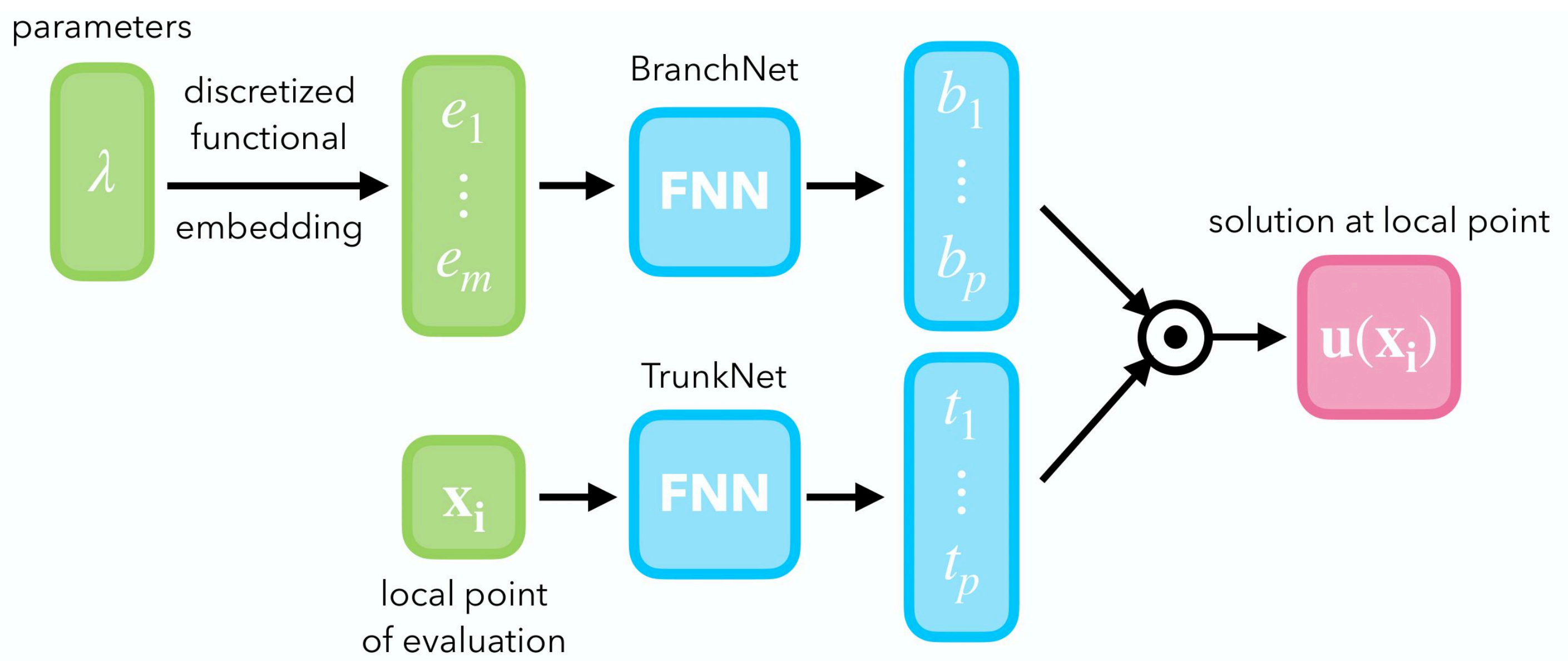
Training objective (MSE):

$$\min_{\theta} \mathcal{L}(\theta) = \frac{1}{n} \sum_{k=1}^n \|\text{FNN}_{\theta}(\lambda_k) - U_k\|^2$$

This architecture is poorly scalable one output per discretization point, and prediction points are fixed.

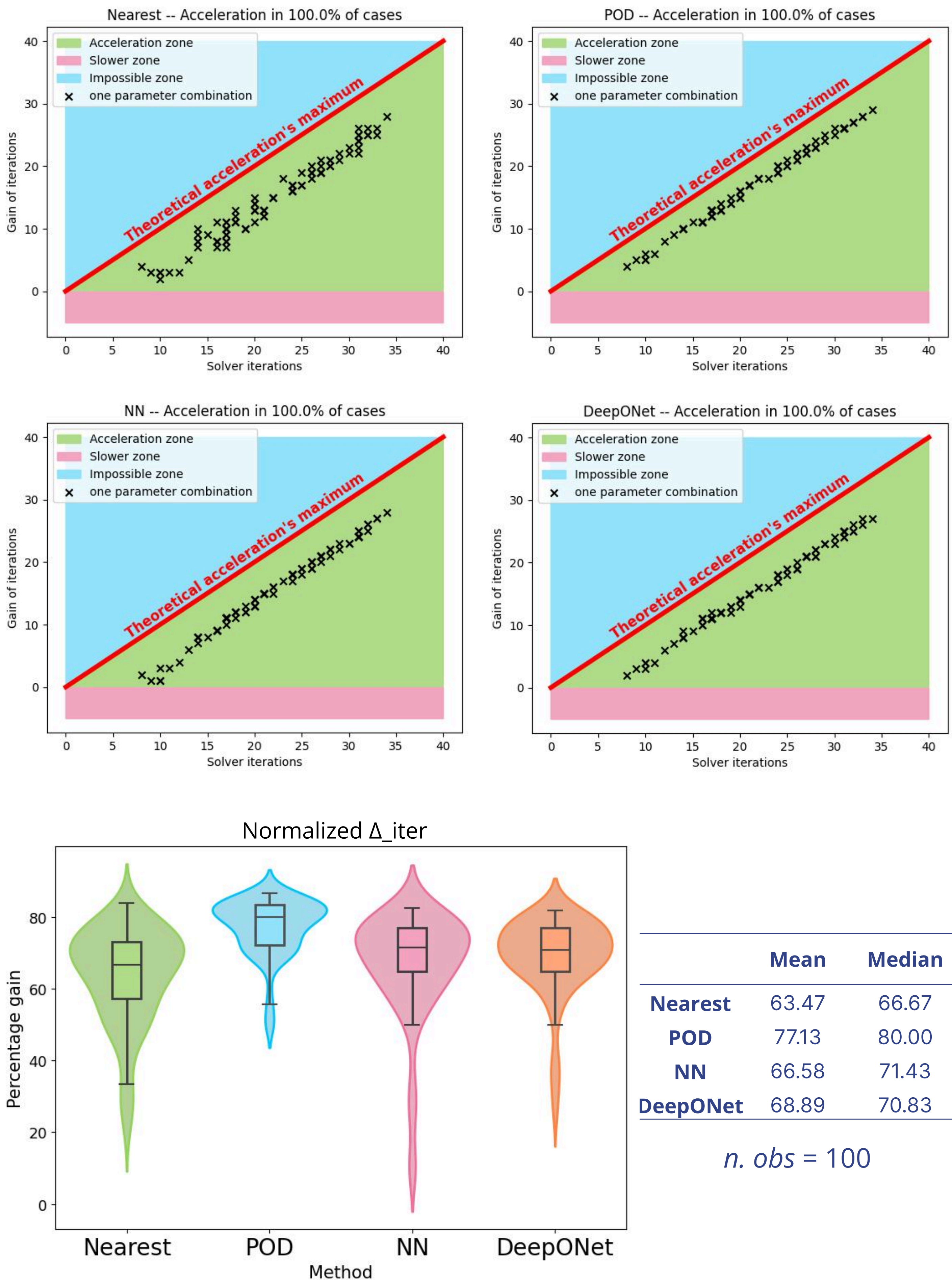
- Deep Operator Networks (DeepONet) [3]

Neural operator architecture to approximate mappings between functions.
Supported by the **Universal Approximation Theorem for operators** [Chen & Chen, 1995].

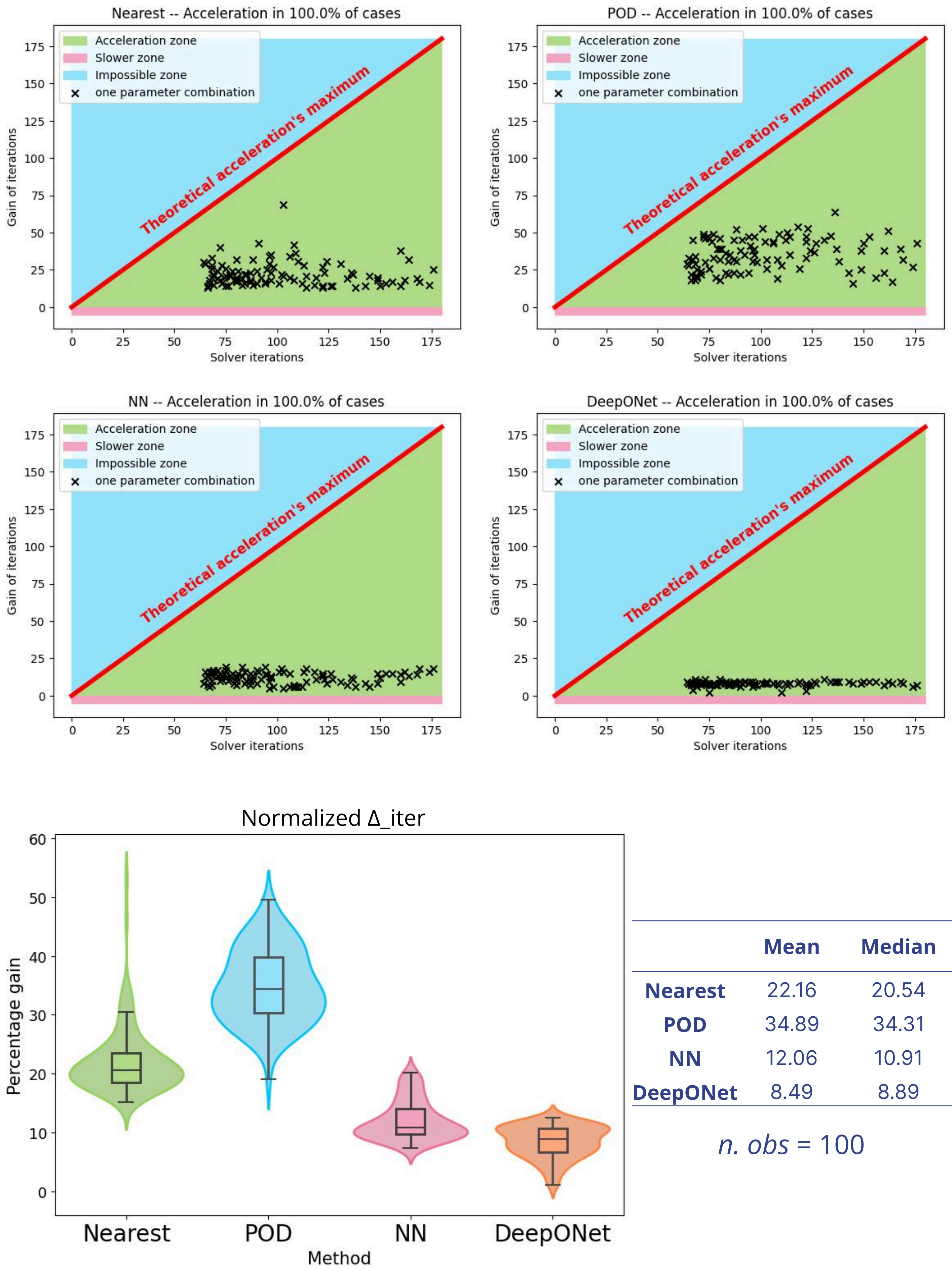


Results

- 1D Nonlinear Poisson equation



- 2D calendering process



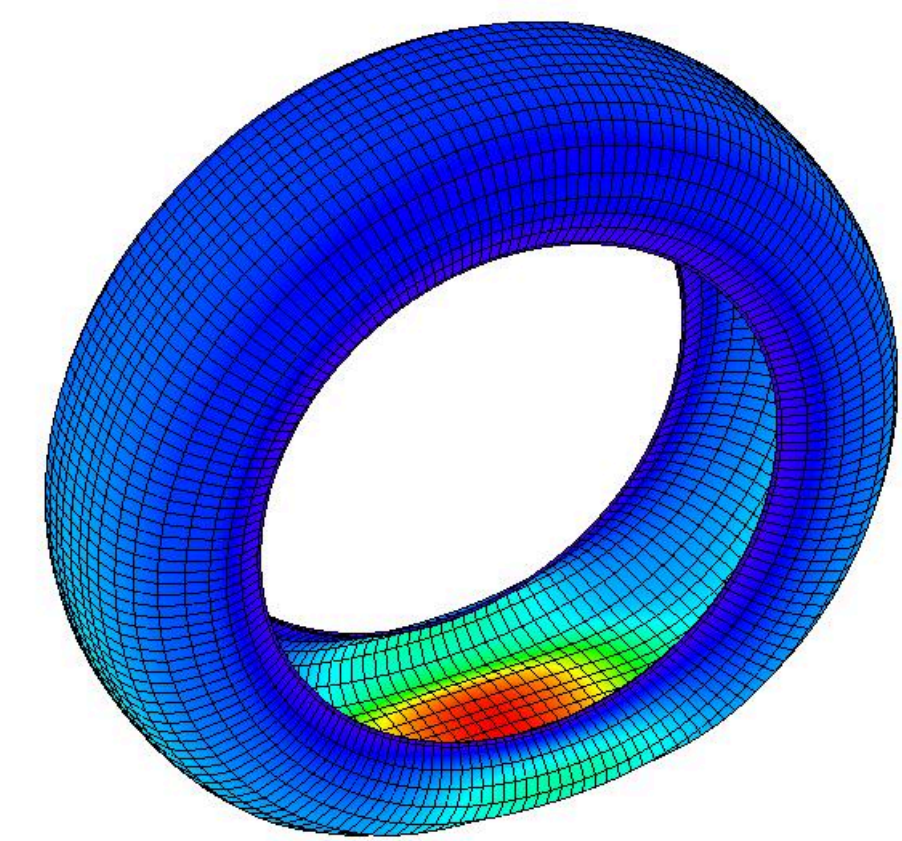
Preliminary conclusions

- All initialization methods perform better than using a constant field.
- POD consistently outperforms all other tested methods.
- In 1D, the iteration gain grows with baseline cost; in 2D the iteration gain is constant.
- Why not PINNs ? Longer training time, inefficient for parametric studies

Future work

- Extension to industrial test cases
- Generalization to geometrical variability

How to assess confidence in surrogate-based initialization ?



References

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